

PHASE 5: Backend Project Setup

Project Title: Academic Credential Verification and Alumni Tracking System

Student: Dusabe Marie Rose

Phase Duration: 21 February – 27 February 2026

Introduction

This phase focused on preparing the backend development environment for the Academic Credential Verification and Alumni Tracking System. The objective was to install the required development tools, initialize the backend project using Spring Boot, configure the database connection, and publish the initial backend project on GitHub.

Completing this phase provided the foundation required for implementing the system's APIs and database operations in the following phases.

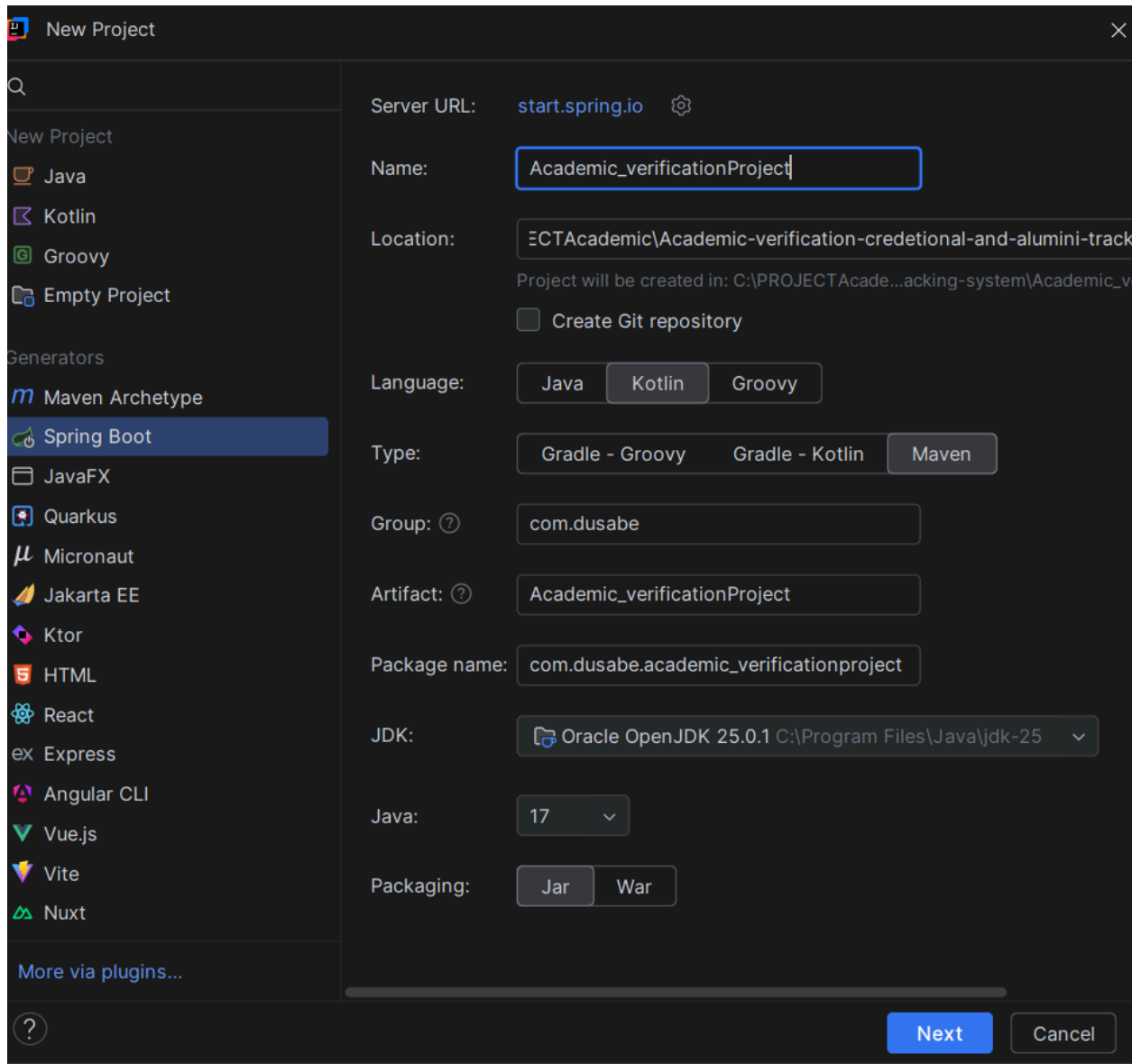
Backend Development Environment Setup

The first step involved preparing the computer with the required development tools used for building Java backend applications. The Java Development Kit was installed to enable Java programming and application execution. A code editor was also installed to allow writing and managing project files, while Git was installed to manage version control of the project.

To confirm that Java was correctly installed, the following command was executed in the terminal:

```
java -version
```

The terminal displayed the installed Java version, confirming that the development environment was ready for backend development.



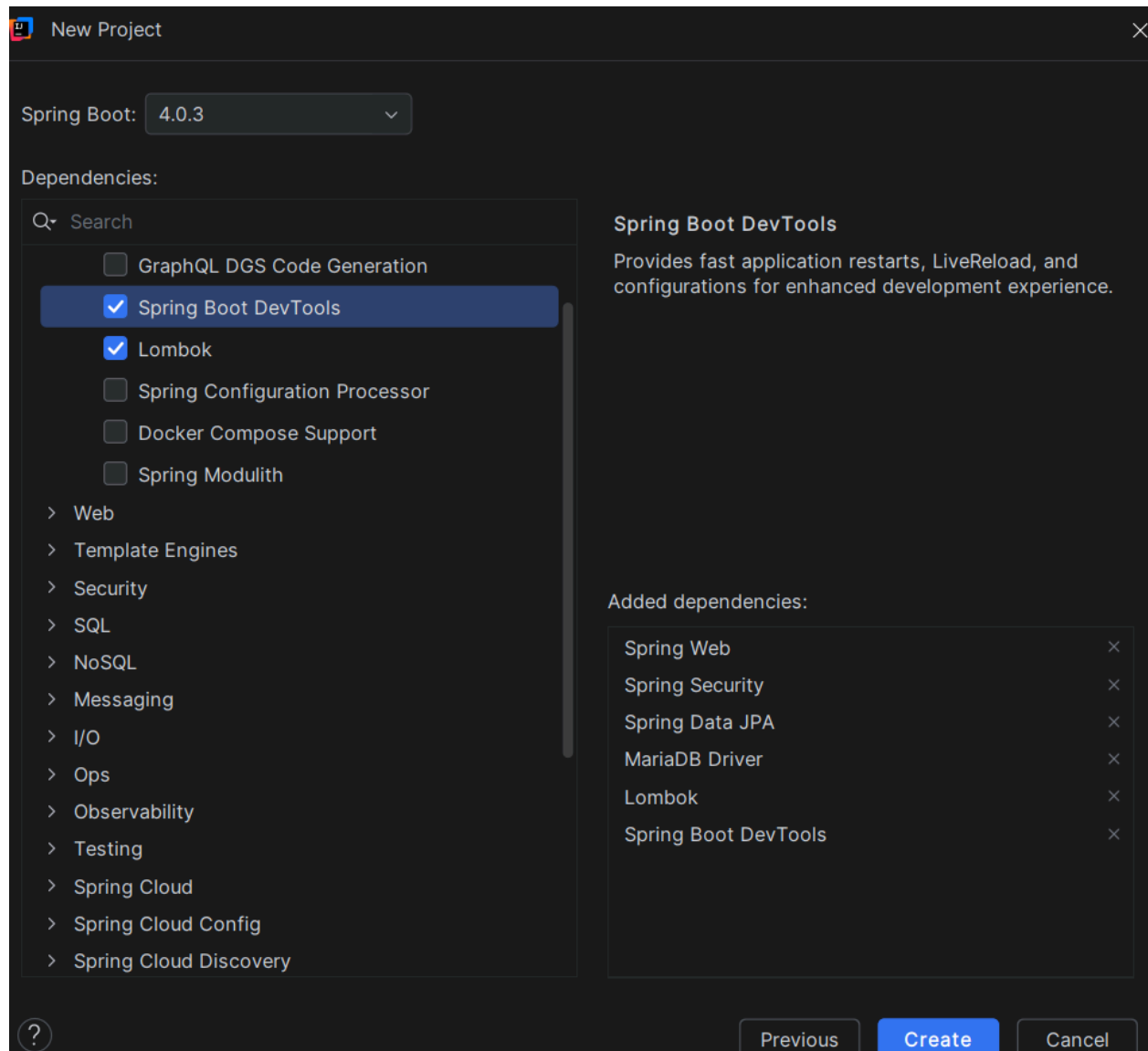
Spring Boot Project Initialization

After setting up the development environment, the backend project was created using Spring Initializr. This tool generates a basic Spring Boot project structure that simplifies backend development.

The project was created using the Maven build tool and the Java programming language. Several dependencies were selected during the initialization process to support web development and database connectivity. These dependencies included Spring Web for creating REST APIs, Spring

Data JPA for database interaction, and the MySQL driver for connecting the application to the database.

Once the project was generated, it was opened in the development environment where further configurations and development tasks were performed.

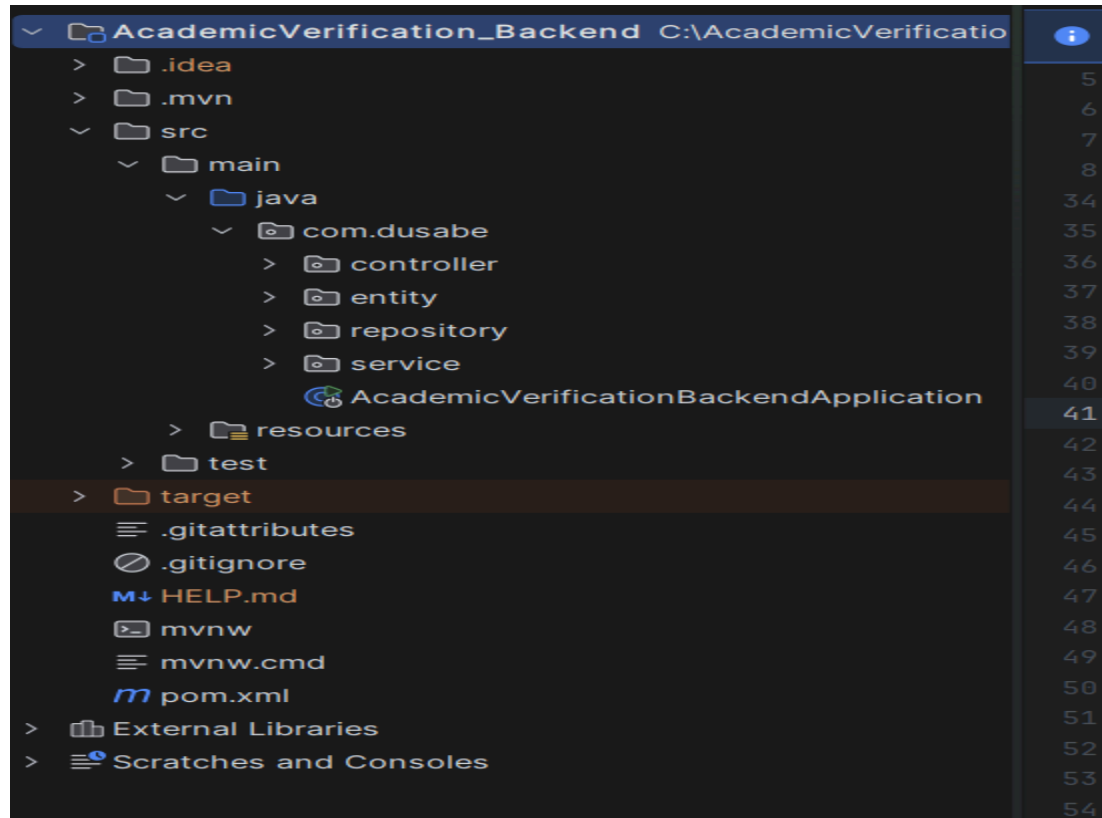


Project Package Structure Creation

After initializing the backend project, the next step involved organizing the project into logical packages. This structure follows the standard architecture used in Spring Boot applications to ensure maintainability and clarity of the code.

The packages created include components responsible for handling different layers of the application such as controllers for handling HTTP requests, services for implementing business logic, repositories for database operations, and entities for representing database tables.

This structure helps maintain a clear separation of concerns between the different parts of the system.

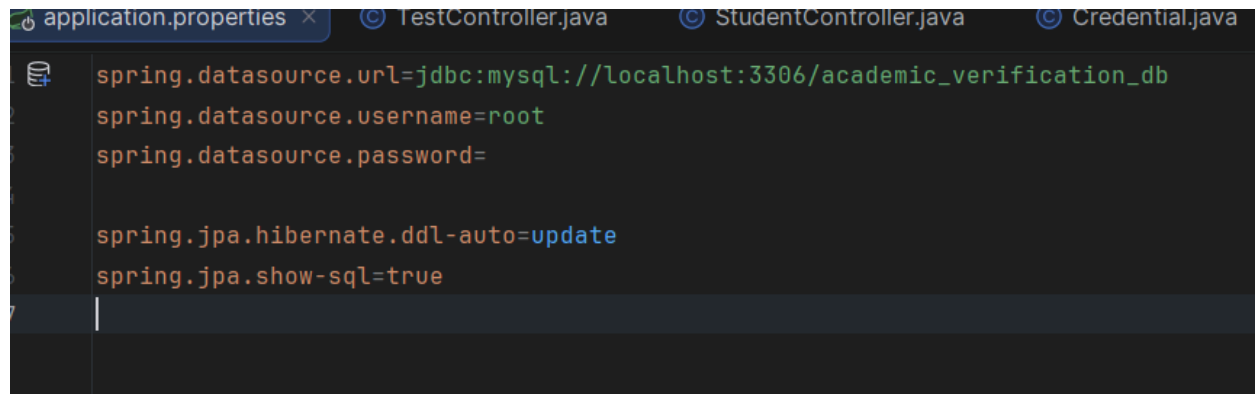


Database Configuration

The backend application was then connected to a MySQL database. This connection allows the application to store and retrieve data related to students, credentials, alumni records, and verification processes.

Database configuration was completed inside the application configuration file where the database URL, username, and password were specified. Additionally, Hibernate settings were configured to automatically update database tables when entity classes are created or modified.

This configuration enabled the backend system to communicate with the database and manage application data efficiently.

A screenshot of an IDE window showing the 'application.properties' file. The tabs at the top are 'application.properties', 'TestController.java', 'StudentController.java', and 'Credential.java'. The code in the editor is as follows:

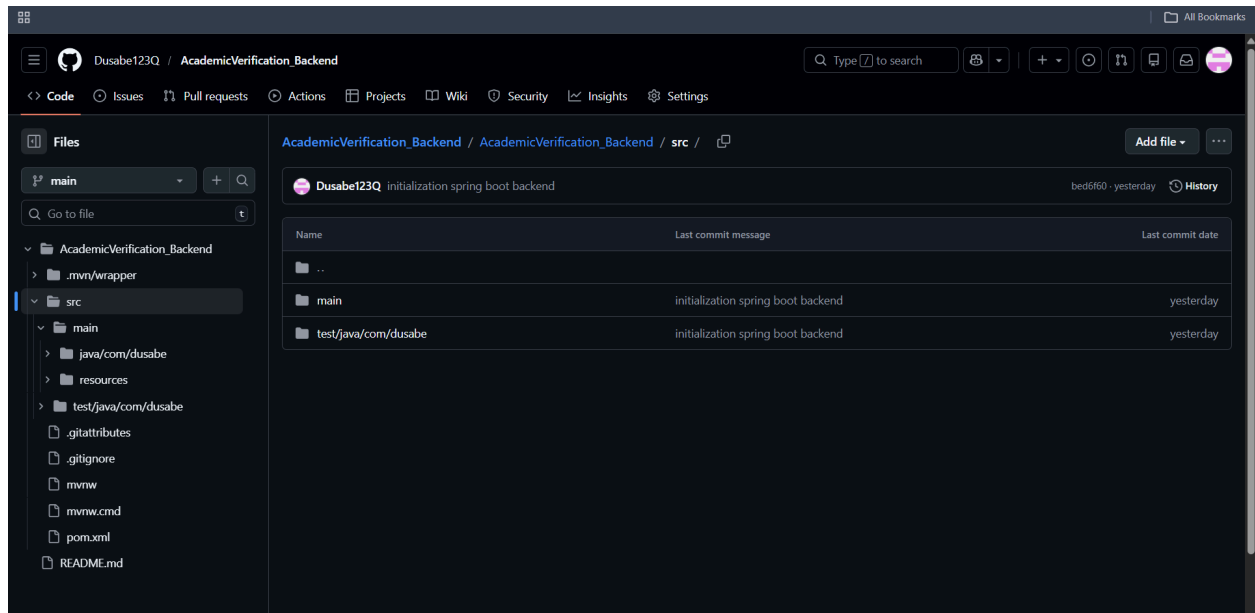
```
spring.datasource.url=jdbc:mysql://localhost:3306/academic_verification_db
spring.datasource.username=root
spring.datasource.password=

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
```

Running and Testing the Application

After configuring the database connection, the backend application was executed to verify that the system was correctly configured. The application was started using the Maven command that runs the Spring Boot server.

Once the application started successfully, it became accessible through the default local server address. The successful startup confirmed that the backend environment and configuration were correctly implemented.



Phase Deliverable

At the end of this phase, the backend system was successfully initialized and prepared for further development. The development environment was configured, the Spring Boot project was created, the database connection was established, and the project was published to GitHub.

These accomplishments provided a solid technical foundation that supports the implementation of backend APIs and system functionality in the following phases.